Hyung Cho

Software Engineer | Coding Agents, Tool Definitions, and LLM Evals

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SUMMARY

Builds full-stack and iOS products by directing AI coding agents, and reviews and verifies every change they make. Writes the tool definitions and judge prompts behind LLM features, then measures them with evals. Spent 10 years as a validation engineer at Zebra Technologies, including two years building C# test automation.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Wrote the tool definitions for a framework-free agent loop on the OpenAI Chat Completions API, where the model picks document search, text-to-SQL, web search, or a sub-agent that reads a full document in its own loop, capped at 5 iterations.
- Traced a 27% false-resolve rate (5% ceiling) to the offline judge's prompt, not the product (0 false resolves in 55 end-to-end runs); compared five judge designs on replayed drafts, and the best cut wrong verdicts from 38 of 280 to 0 of 300.
- Found the live answer gate's yes/no judge passed an off-target reply 10 of 20 times; on 85 stored drafts, a stricter rubric let 1 of 66 should-escalate drafts through against 15 of 66 for a stronger model, at about $0.00007 more per reply.
- Built a multi-user RAG chat app: a FastAPI backend on the raw OpenAI SDK with no LLM framework, Supabase auth and storage, streamed responses, and a React and TypeScript UI with per-chunk source badges and LangSmith traces.
- Built a 60-question retrieval benchmark that scores vector, keyword, and hybrid search on recall@k, MRR, and nDCG@5 and posts the difference from main on pull requests; it flagged a 0.510 recall@5 drop on a deliberately broken change.

Rep Today, native iOS workout app, pre-release

- Ran development as 196 agent-authored pull requests over about 15 weeks, 180 of them with a written risk assessment; later ones separate scenarios validated live in the app from those covered only by tests.
- Built a premium Coach chat that sends the user's message through a JavaScript Cloudflare Worker proxy to OpenAI (store: false, capped output, 30-second timeout), with App Attest and subscription checks on the server.
- Instrumented 13 pre-registered funnel events through a Convex HTTP endpoint with opt-out consent and rate limits, and verified accepted, rejected, rate-limited, and opted-out behavior in production checks.

CreativeBridge, K-12 AI story-writing iOS app, internal TestFlight builds

- Built a React Native and Expo iOS app with Claude Code (311 of 426 commits co-authored by Claude).
- Added on-device speech-to-text with whisper.rn so audio from users under 13 never leaves the device, with an opt-in cloud path for grades 9-12 behind a disclosure and logged consent.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Fine-tuned open-source LLMs (LLaMA, Falcon) and iterated on chatbot prompts with an evaluation workflow.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Built C# validation tooling that replaced manual test workflows and cut regression test cycles by about 30%.
- Led product validation for 6 shipped hardware products, writing test plans, acceptance criteria, and final reports.
- Owned SDK validation on Windows, macOS, and Linux; isolated failures across hardware, firmware, and software.

SKILLS

Coding agents and prompts: AI coding agents, Claude Code, Tool calling, Agent loops, Prompt engineering
Evals and testing: LLM evaluation, LLM-as-judge, Retrieval metrics, Test automation, Regression testing, GitHub Actions
Full stack: Python, FastAPI, JavaScript, TypeScript, React, Cloudflare Workers, PostgreSQL, Supabase
APIs and writing: OpenAI API, Anthropic API, LLM integration, RAG, whisper.rn, Technical writing

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo
FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses